Sequence & Progression: Formula Sheet

1. Arithmetic progression

n^{th} term: $T_n = a + (n-1)d$

Sum:

$$S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a + l)$$

Number of terms:

$$n = \frac{l - a}{d} + 1$$

From S_n: $T_n = S_n - S_{n-1}$

Condition: a, b, c in AP $\Leftrightarrow 2b = a + c$

Equidistant terms: $T_k + T_{n-k+1} = a + l$

Ratio of sums:

$$\frac{T_k}{T'_k} = \frac{S_{2k-1}}{S'_{2k-1}}$$

Common terms of two APs: $d = \text{lcm}(d_1, d_2)$

2. Arithmetic mean

AM:

$$A = \frac{a_1 + \cdots + a_n}{n}, \quad \frac{a+b}{2}$$

n AMs:

$$d = \frac{b-a}{n+1}, \; A_k = a + kd$$

Sum of n AMs:

$$\sum A_k = n \cdot \frac{a+b}{2}$$

3. Geometric progression

n^{th} term: $T_n = ar^{n-1}$

Sum:

$$S_n = \frac{a(r^n - 1)}{r - 1} \; (r \neq 1)$$

Infinite sum:

$$S_\infty = \frac{a}{1-r}, \; |r| < 1$$

Condition: a, b, c in GP $\Leftrightarrow b^2 = ac$

Sum of squares (infinite):

$$\frac{a^2}{1 - r^2}$$

Recurring decimal:

$$0.\overline{pq} = \frac{pq}{99}$$

$a_{n+1} = pa_n + q$:

$$a_{n+1} + k = p(a_n + k), \; k = \frac{q}{p-1}$$

Triangle sides a, ar, ar^2:

$$1 < r < \frac{1 + \sqrt{5}}{2}$$

4. Geometric mean

GM:

$$G = (a_1 \cdots a_n)^{1/n}, \quad \sqrt{ab}$$

n GMs:

$$r = \left(\frac{b}{a}\right)^{\frac{1}{n+1}}, \; G_k = ar^k$$

Product of n GMs: $(ab)^{n/2}$

5. Harmonic progression and mean

HP:

$$T_n = \frac{1}{a + (n-1)d}$$

Condition:

$$b = \frac{2ac}{a + c}$$

HM:

$$H = \frac{n}{\sum \frac{1}{a_i}}, \quad \frac{2ab}{a+b}$$

n HMs:

$$\frac{1}{H_k} = \frac{1}{a} + kd, \; d = \frac{\frac{1}{b} - \frac{1}{a}}{n+1}$$

Sum of reciprocals of n HMs:

$$n \cdot \frac{\frac{1}{a} + \frac{1}{b}}{2}$$

6. Means and inequalities

AM-GM: $\frac{\sum a_i}{n} \geq (\prod a_i)^{1/n}$ (equality: all equal)

Chain:

$$\text{RMS} \geq \text{AM} \geq \text{GM} \geq \text{HM}$$

Two numbers: $G^2 = AH$

Equation with roots a, b: $x^2 - 2Ax + G^2 = 0$

Max of $\prod a_i^{p_i}$ with $\sum a_i = s$: $a_i \propto p_i$

7. Arithmetico-geometric progression

General term: $T_n = [a + (n-1)d]\, r^{n-1}$

Infinite sum:

$$S_\infty = \frac{a}{1-r} + \frac{dr}{(1-r)^2}$$

n terms:

$$S_n = \frac{a}{1-r} + \frac{dr(1 - r^{n-1})}{(1-r)^2} - \frac{[a + (n-1)d]r^n}{1-r}$$

$$1 + 2x + \cdots + nx^{n-1} = \frac{1 - (n+1)x^n + nx^{n+1}}{(1-x)^2}$$

8. Standard sums (type 1)

$$\sum n = \frac{n(n+1)}{2}$$

$$\sum n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum n^3 = \left[\frac{n(n+1)}{2}\right]^2$$

Alternating squares:

$$\sum_{k=1}^{n}(-1)^{k-1}k^2 = (-1)^{n-1}\frac{n(n+1)}{2}$$

Products two at a time:

$$\frac{(\sum a_i)^2 - \sum a_i^2}{2}$$

$$\sum r(r+1) = \frac{n(n+1)(n+2)}{3}$$

9. Types 2, 3 and 4

Type 2:

$$T_n = u_1 + \sum_{r=2}^{n}(u_r - u_{r-1})$$

Method of difference:

$$T_r = f(r) - f(r-1) \Rightarrow S_n = f(n) - f(0)$$

Type 3:

$$\frac{1}{n(n+1)(n+2)} = \frac{1}{2}\left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right]$$

Type 3 (step d):

$$\frac{1}{a_n \cdots a_{n+r-1}} = \frac{1}{(r-1)d}\left[\frac{1}{a_n \cdots a_{n+r-2}} - \frac{1}{a_{n+1} \cdots a_{n+r-1}}\right]$$

Type 4:

$$\sum n(n+1)(n+2)(n+3) = \frac{n(n+1)(n+2)(n+3)(n+4)}{5}$$

$$\frac{r}{(r+1)!} = \frac{1}{r!} - \frac{1}{(r+1)!}, \quad \frac{1}{\sqrt{k}+\sqrt{k+1}} = \sqrt{k+1} - \sqrt{k}$$

10. Roots in progression and counting

Cubic, roots in AP: middle root $= -\frac{b}{3a}$

Cubic, roots in GP: (middle root)$^3 = -\frac{d}{a}$

Numbers in a range divisible by a or b:

$$S(a) + S(b) - S(\text{lcm}(a,b))$$

n^{th} group of $1; 2, 3, 4; 5, \ldots, 9; \ldots$: sum $(n-1)^3 + n^3$

Bouncing ball: total distance $H + \frac{2He}{1-e}$